# CryspLight full simulation: working plan

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## 1 Scope

Build the full simulation specified in `fullsim_design.tex` and compare it with the bench measurements: BGO and CsI(Tl) at ambient temperature with the Citiroc 1A and with the DT2745 digitizer (data taking the week of 20 October 2026), BGO at 196 K with the digitizer afterwards, Radioroc2 when the board arrives. This document tracks what is planned, in progress and done, the decisions taken and the questions open. It is edited at every step; the physics stays in the design document.

## 2 Principles

- The pedagogical simulation is the git tag `pedsim` and the branch `main`; the full simulation grows on the branch `fullsim`. Nothing is replaced: every feature is gated by a configuration key and the frozen-output test proves the pedagogical path unchanged.
- Repository layout for the full simulation: `docs_fullsim/` (documents), `fullsim_runs/<campaign>/` (simulation and analysis configurations plus the run log of each measurement campaign), `fullsim_analysis/` (Python comparison layer, committed figures under `results/<campaign>/`), engine additions in `src/`, the electronics emulation in `src/frontend/`, tests in `test/`. Simulation output in the gitignored `output/<tag>/`. Measured data outside the repository, referenced by path from the campaign configuration.
- Documents are written before the code they describe; results are produced by tracked scripts and committed with their images; every reported result states where it lives.

## 3 Phases and tasks

### Phase 0 — before and during the ambient runs

| id | task | status | date |
|---|---|---|---|
| 0.1 | Tag `pedsim`, branch `fullsim`, datasheets committed | done | 16 Oct |
| 0.2 | Design document and this plan | done | 16 Oct |
| 0.3 | Overvoltage recipe for the bench (`BGOTSCAN/docs/overvoltage_recipe.tex`) | done | 16 Oct |
| 0.4 | Frozen-output test of the pedagogical path | planned | |
| 0.5 | Readout layout: active rectangles, dead-area reflection, overhang; parity and layout tests | planned | |
| 0.6 | Grease and resin window: second Fresnel interface; window test | planned | |

| id | task | status | date |
|---|---|---|---|
| 0.7 | Materials: cold BGO file (2.7 µs), ambient BGO file; two-component CsI(Tl) timing | planned | |
| 0.8 | $^{22}$Na point source with both lines and coincidence summing; source test | planned | |
| 0.9 | Photoelectron records `pes.h5`; re-binning test | planned | |
| 0.10 | SiPM device model (gain, PDE, crosstalk, afterpulsing, dark counts, pulse) as a module with unit tests | planned | |
| 0.11 | Citiroc chain: shapers, noise, trigger, 64 peak detectors, hold, ADC, saturation; unit tests (the stored fractions 0.39/0.15/0.07 are the first test) | planned | |
| 0.11b | Digitizer chain on the avalanche train: gate and baseline weights, integrated noise, sampled-amplitude trigger with majority, waveform subsample; unit tests against the analytic noise and a brute-force waveform on a few events | planned | |
| 0.12 | Common event layout writer (emulation) and the converter skeleton for measured files | planned | |
| 0.13 | First emulated campaign: BGO ambient and CsI(Tl) ambient at 2, 2.7, 3, 4 and 5 V overvoltage, with both readouts; tables of photoelectrons per channel, brightest-channel tail, stored fraction, resolution floor, trigger margin; for the digitizer the trigger efficiency and dark-trigger rate against threshold and the resolution against gate length | planned | |
| 0.14 | Overvoltage scan figure and numbers delivered to the bench | planned | |

**Phase 1 — when the ambient data exist**

| id | task | status | date |
|---|---|---|---|
| 1.1 | Converters from the bench files to the common layout, with the run log: Citiroc charges; digitizer waveforms reduced to baseline, gated charge and threshold crossings (from the Antarctica analysis code) | planned | |
| 1.2 | Single-photoelectron gains and breakdown per channel; pedestals and noise term; dark rates | planned | |
| 1.3 | Spectra: 511, 1275, sum peak; resolution and its decomposition; 1275 to 511 ratio | planned | |
| 1.4 | Light yield in photoelectrons and in photons per MeV through the simulated collection | planned | |
| 1.5 | Maps: occupancy, mean charge, seed-channel photopeak; fit of the dead-area reflectivity and check of the roughness | planned | |
| 1.6 | Brightest-channel distribution and saturation in CsI(Tl) | planned | |
| 1.7 | Emulation rerun with the measured noise, gains and settings; side-by-side figures | planned | |
| 1.8 | Note section per campaign in the results document | planned | |

**Phase 2 — accuracy and the next setups**

| id | task | status | date |
|---|---|---|---|
| 2.1 | Spectral sampling: emission spectra, $n(\lambda)$, PDE($\lambda$); spectral test | planned | |
| 2.2 | Cold BGO campaign at 196 K: breakdown measured in situ, emulation, comparison; warm-to-cold yield ratio against the quenching curve | planned | |
| 2.3 | Radioroc2 chain and its campaign | planned | |
| 2.4 | Gamma stage: K-shell fluorescence and escape | planned | |
| 2.5 | Position reconstruction on emulated maps: how much of the `pedsim` CNN resolution survives each front end | planned | |
| 2.6 | Design document version 1.0 with the measured parameters | planned | |

# 4 Campaigns

| campaign | configuration | readout |
|---|---|---|
| `2026-10_ambient` | BGO ambient, CsI(Tl) ambient | Citiroc 1A ×2 and DT2745, in sequence on th |
| `2026-xx_digitizer_cold` | BGO 196 K | DT2745 |
| `2026-xx_radioroc` | all three | Radioroc2 |

# 5 Decisions

**16 Oct 2026** Full simulation as a subproject of CryspLight, not a client package: most of the work is engine work under the parity test; one history, one test suite.

**16 Oct 2026** Additive only. Tag `pedsim`; frozen-output test; gated features; separate files for the photoelectron records and the emulated events; pedagogical configurations and outputs untouched.

**16 Oct 2026** Three-stage pipeline with files between stages, so that the front end can be retuned against data without rerunning the optics.

**16 Oct 2026** Grease ($n = 1.42$) in every configuration; not a parameter.

**16 Oct 2026** Cold BGO decay time 2.7 µs from the Antarctica runs, in a new material file; the Ding file stays for the pedagogical runs.

**16 Oct 2026** The Antarctica runs are not simulated (dead channels, noisy chain); the first comparison is the Citiroc ambient campaign.

**16 Oct 2026** Overvoltage set from the measured breakdown, 3.0 V for BGO and 2.7 V for CsI(Tl); recipe in `BGOTSCAN/docs/overvoltage_recipe.tex`.

**16 Oct 2026** Measured and emulated events share one HDF5 layout; the bench files are converted to it as the first analysis step.

**16 Oct 2026** No waveform-digitiser emulation. *Reversed the same day*, see below.

**16 Oct 2026** The DT2745 digitizer is a third front end of the simulation, emulated on the avalanche train (gated charge with baseline subtraction, integrated noise, sampled-amplitude trigger) rather than through waveforms. Reason: the Citiroc at 87.5 ns stores 0.07 of the photoelectrons of cold BGO, so the cold runs are taken with the digitizer, and the ambient runs with both readouts give a parameter-free test of the stored fraction. The waveform

processing of the measured data (baseline, single-photoelectron search) stays on the data side.

**16 Oct 2026** Digitizer noise: nominal (4 counts rms, white, independent channels) until the new runs; the low-frequency and group-correlated terms are added then as a measured input.

**16 Oct 2026** Afterpulsing 1% at 3 V, scaling with the square of the overvoltage, 100 ns; not resolvable with the Citiroc.

**16 Oct 2026** Intrinsic resolution 5% FWHM at 511 keV for both crystals; the non-proportionality curve of BGO (Gentile et al.) as a later refinement; Moszynski et al. to be checked.

**16 Oct 2026** Ambient digitizer runs at gain 1 and 43 V (5 V of overvoltage), the simplest setting for the bench; threshold 60 counts, majority 2, record 10 µs with 1 µs pre-trigger.

## 6 Open questions

- Reflectivity of the tile surface between channels (default diffuse 0.5; to fit on the occupancy maps).
- BGO light yield at ambient and at 196 K for this crystal; BGO intrinsic resolution.
- Citiroc settings of the ambient runs (shaping time, gains, hold delay, thresholds, trigger definition in the FPGA) and the ADC; to be written in the run log.
- Angular dependence of the PDE through the resin from grease.
- Citiroc shaper transfer function and noise spectrum; whether a single-photoelectron waveform can be probed on the test board.
- Temperature of the SiPMs in the cold run (resolved by the measured breakdown; a thermometer on the SiPM board would still help).
- Whether the digitizer gain stage lowers the noise per photoelectron (baseline runs at gains 1, 2, 4, 8 with the SiPMs unbiased); decides the overvoltage of the cold runs.
- Whether the Radioroc2 at 1.8 µs shaping (0.55 of the photoelectrons of cold BGO) is usable in the cold, or a less cold point is preferable.
- The format of the WaveDump2 output files and the event-building of the 64 channels, for the converter.

## 7 Log

**16 Oct 2026** Project assessed; datasheets read (Citiroc 1A, Radioroc2, S14160/S14161); Antarctica draft read; tag and branch; design document v0.1 and this plan; overvoltage recipe written.

**16 Oct 2026** Design document reviewed with JJ: measured data (one high-gain charge and the hit register), afterpulsing, the stored fraction recomputed with the avalanche pulse (0.39/0.15/0.07), digitizer manual (UM8717) read and the DT2745 added as a third front end; digitizer section added to the overvoltage recipe; trigger thresholds for the Citiroc and the digitizer estimated for the ambient runs.